

Exercises for MATH-4100

Summer 2025

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- 1) Given two vectors $\mathbf{a} = (1, 1, 2)$ and $\mathbf{b} = (3, 0, -1)$, find a third vector $\mathbf{c} \in \mathbb{R}^3$ that orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.
- 2) Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ be such that $\mathbf{a} + \lambda \mathbf{b} \neq \mathbf{0}$ for all $\lambda \in \mathbb{R}$. Show that if both $\mathbf{c}$ and $\mathbf{d}$ are othogonal to $\mathbf{a}$ and $\mathbf{b}$ then $\mathbf{c}$ and $\mathbf{d}$ must be parallel.
- 3) Prove the parallelogram law: $\|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 = 2(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2)$ for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
- 4) Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Show that:
 - a. $\mathbf{x}^T \mathbf{y} = 0$ if and only if $\|\mathbf{x}\| \leq \|\mathbf{x} + a\mathbf{y}\|$ for all $a \in \mathbb{R}$
 - b. If $\|\mathbf{x}\| = \|\mathbf{y}\|$ then $\|a\mathbf{x} + b\mathbf{y}\| = \|b\mathbf{x} + a\mathbf{y}\|$ for all $a, b \in \mathbb{R}$

- 5) Let a, b , and c be lenghts of the sides of a triangle and let d be the length of the line segment from the midpoint of the side of length c to the opposite vertex. Use inner products to prove that

$$a^2 + b^2 = \frac{1}{2}c^2 + 2d^2$$

- 6) Let $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ be a set of vectors in $\mathbb{R}^n$ such that $\mathbf{x}_i^T \mathbf{x}_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$. Show that if $\mathbf{v} = \sum c_i \mathbf{x}_i$ then $c_i = \mathbf{x}_i^T \mathbf{v}$. (Note that the vectors $\mathbf{x}_i$ are said to be orthonormal)

- 7) Let V be an inner product space and let $\{e_1, \dots, e_n\}$ be an orthonormal basis for V , that is $\langle e_i, e_j \rangle = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$. Show that for any $x, y \in V$ with $x = \sum_{i=1}^n x_i e_i$ and $y = \sum_{i=1}^n y_i e_i$ we have that $\langle x, y \rangle = \sum_{i=1}^n x_i y_i$.

- 8) Show that the following subsets of $\mathbb{R}^n$ are subspaces:

- a. $W_1 = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = \sum c_i \mathbf{v}_i\}$ where $\mathbf{v}_i \in \mathbb{R}^n$ for each i . (This is the set of linear combinations of the vectors $\mathbf{v}_i$)
- b. $W_2 = \{\mathbf{y} \in \mathbb{R}^n : \mathbf{y} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\}$ where $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$.

- 9) Show that the following subsets of $\mathbb{R}^n$ are not subspaces:

- a. $S = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{b}\}$ where $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ and $\mathbf{b} \neq \mathbf{0} \in \mathbb{R}^n$.
- b. Any non-empty, finite subset of $\mathbb{R}^n$. (Clarification: by non-empty I mean containing at least one non-zero vector)

- 10) Let V be a vector space over F and let U_1 and U_2 be subspaces. Prove that $U_1 \cap U_2$ is also a subspace.

- 11) Show by example that the union of two subspaces need not be a subspace.

- 12) Let V be a vector space over F and let S and T be subspaces such that $U_1 \cap U_2 = \emptyset$. Show that for any $x \in S + T$, there exists unique $s \in S$ and $t \in T$ such that $x = s + t$.
- 13) Given $U_1 = \{(x_1, 0, x_3) \in \mathbb{R}^3 : 4x_1 + 2x_3 = 0\}$ and $U_2 = \{(0, x_2, x_3) \in \mathbb{R}^3 : x_2 = 5x_3\}$, compute $U_1 + U_2$.
- 14) Let $W_1 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 = 3x_2\}$ and $W_2 = \{(t, -3t, 0) : t \in \mathbb{R}\}$ be two subspaces of $\mathbb{R}^3$. Show that $\mathbb{R}^3 = W_1 \oplus W_2$.
- 15) Determine whether or not the following three vectors in $\mathbb{R}^3$ linearly independent and describe their span

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

- 16) Let V be a finite dimensional vector space and let $\mathcal{B}$ and $\mathcal{B}'$ be two bases. Show that $|\mathcal{B}| = |\mathcal{B}'|$, i.e. show that both sets have the same number of elements.
- 17) Let $\mathbb{P}_n$ be the space of all polynomials over $\mathbb{R}$ of degree at most n . Show that the subset $S = \{f \in \mathbb{P}_n : f'(1) = 0\}$ is a subspace of $\mathbb{P}_n$ and find a basis.
- 18) Let V be a vector space and let L be any linear operator on V , i.e. $L \in \mathcal{L}(V, V)$. Prove that the subset $S = \{y \in V : y = L[x] \text{ for some } x \in V\}$ is a subspace of V .
- 19) Let $T \in \mathcal{L}(V, W)$ be a linear transform and define $\ker(T) = \{x \in V : T(x) = \mathbf{0}\}$. Prove that T is injective if and only if $\ker(T) = \mathbf{0}$. ($\ker$ =kernel)
- 20) Let V be finite dimensional and let $T \in \mathcal{L}(V, V)$. Show that T is surjective if and only if $\ker(T) = \mathbf{0}$. Does this remain true if V is infinite dimensional?

21) Given $A = \begin{bmatrix} 1 & 2 & 4 \\ -2 & -3 & 2 \\ 2 & 2 & -1 \end{bmatrix}$, and $B = \begin{bmatrix} 3 & -2 & 6 \\ 1 & 1 & 0 \\ -1 & 3 & 2 \end{bmatrix}$ compute AB and BA

- 22) Find the null space and range of the following matrices

a. $A = \begin{bmatrix} 1 & 2 & 0 \\ 6 & 5 & 1 \\ -3 & 1 & -1 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & -2 & 1 & 0 & 3 \\ -3 & 6 & 0 & 2 & -1 \\ -2 & 4 & 3 & 1 & 0 \end{bmatrix}$

- 23) $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times l}$. Prove that $\text{null}(B) \subseteq \text{null}(AB)$ and $\text{range}(AB) \subseteq \text{range}(A)$. (This demonstrates that $\text{nullity}(B) \leq \text{nullity}(AB)$ and $\text{rank}(AB) \leq \text{rank}(A)$.)
- 24) Let A and B be invertible $n \times n$ matrices. Prove that
- a. $(AB)^{-1} = B^{-1}A^{-1}$
- b. $(A^{-1})^T = (A^T)^{-1}$

25) Given $A = \begin{bmatrix} 1 & 2 & 4 \\ -2 & 0 & 2 \\ 2 & 2 & -1 \end{bmatrix}$, compute the following

- a. A^2
- b. A^T
- c. A^{-1}

26) Prove that if $A \in \mathbb{R}^{m \times n}$ where $n \neq m$, then A is not invertible.

27) Let $A \in \mathbb{R}^{m \times n}$ where $m < n$ be such that $\text{rank}(A) = m$. Show that AA^T is invertible, but $A^T A$ is not.

28) Show that every matrix can be decomposed into a symmetric and skew symmetric matrix. I.e, for any matrix A we can find matrices B and C such that $B^T = B$, $C^T = -C$, and $A = B + C$. (This corresponds with the property that any function can be written as the sum of an even and odd function.)

29) Find the general solution to $A\mathbf{x} = \mathbf{b}$ given that

$$A = \begin{bmatrix} 1 & 0 & 2 & 4 \\ 2 & 1 & 1 & 3 \\ 3 & 1 & 3 & 7 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 2 \\ 6 \\ 8 \end{bmatrix}$$

30) Let S be the subspace of $\mathbb{R}^4$ given by $S = \{(x_1, x_2, x_3, x_4) : x_1 - 2x_2 + 4x_3 - 3x_4 = 0\}$

- a. Find a basis for S and $S^\perp$
- b. Find a matrix M such that $S^\perp = \text{null}(M)$

31) Let S be the subspace of $\mathbb{P}_3$ given by $S = \{f \in \mathbb{P}_3 : f(1) = f'(1) = 0\}$. Find bases for S and $S^\perp$. (You may assume an inner product where the standard basis is orthonormal)

32) Find a basis for $S^\perp$ from above using the typical inner product, $\langle f, g \rangle = \int_0^1 fg$

33) Prove that given any matrix A , $\text{range}(A)^\perp = \text{null}(A^T)$ and $\text{range}(A^T)^\perp = \text{null}(A)$.

34) Compute $\text{range}(A^T)$ and $\text{null}(A^T)$

$$\begin{aligned} \text{a. } A &= \begin{bmatrix} 1 & 2 & 0 \\ 6 & 5 & 1 \\ -3 & 1 & -1 \end{bmatrix} \\ \text{b. } A &= \begin{bmatrix} 1 & -2 & 1 & 0 & 3 \\ -3 & 6 & 0 & 2 & -1 \\ -2 & 4 & 3 & 1 & 0 \end{bmatrix} \end{aligned}$$

35) Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{m \times l}$. Show that $\text{range}(A)$ is orthogonal to $\text{range}(B)$ if and only if $A^T B$ is the zero matrix

36) Recall that a directed graph consists of nodes and directed edges connecting two nodes. We define a connected component of the graph to be a collection of nodes that share an edge with at least one other node in the collection. Let A be the incidence matrix for the directed graph G . Show that $\text{nullity}(A)$ is equal to the number of connected components of G .

37) Let

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{bmatrix}$$

- Draw a directed graph having A as its incidence matrix
- Find bases for the four fundamental subspaces; $\text{range}(A)$, $\text{range}(A^T)$, $\text{null}(A)$, and $\text{null}(A^T)$

38) Let $\{\mathbf{e}_i\}$ be the standard basis for $\mathbb{R}^2$. Let T be the linear transformation on $\mathbb{R}^2$ defined as follows

$$\begin{aligned} T\mathbf{e}_1 &= \mathbf{e}_1 + 2\mathbf{e}_2 \\ T\mathbf{e}_2 &= 3\mathbf{e}_1 - 3\mathbf{e}_2 \end{aligned}$$

- Find $[T]_{\{\mathbf{e}_i\}}$ the matrix representation of T with respect to the standard basis.
- Let $\{\mathbf{v}_1, \mathbf{v}_2\}$ be another basis for $\mathbb{R}^2$ where $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$. Find $[T]_{\{\mathbf{v}_i\}}$, the matrix representation of T with respect to the basis $\{\mathbf{v}_i\}$ and use it write $T \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ as a linear combination of the basis vectors, $\mathbf{v}_i$.

39) Let T be a linear transformation on $\mathbb{R}^n$ such that $T\mathbf{e}_i = \mathbf{v}_i$ for $i = 1, \dots, n$.

- Find $[T]_{\{\mathbf{e}_i\}}$ the matrix representation of T w.r.t the standard basis
- Now assume that for linearly independent vectors $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ we have that $T\mathbf{w}_i = \lambda_i \mathbf{w}_i$ for each i . Show that the matrix representation of $[T]_{\{\mathbf{w}_i\}}$ is a diagonal matrix.
- Finally state the change of basis matrix from $\{\mathbf{e}_i\}$ to $\{\mathbf{w}_i\}$ and find a relationship between each of the previously mentioned matrices.

40) Let $L : \mathbb{P}_3 \rightarrow \mathbb{P}_3$ be defined by $L[f] = D[(2x + 3)f]$ where $D = d/dx$ is the differential operator.

- Show that L is a linear operator
- Find $[L]$ with respect to the standard basis $\{1, x, x^2, x^3\}$ and show that L is onto. (Think dimensionality)
- Use the above result to solve the equation $L[f] = x^2$. You may use a CAS here.

41) Let $T : \mathbb{P}_2 \rightarrow \mathbb{R}_3$ be defined by $T(f) = (f(-3), f(2), f(5))$.

- Show that T is a linear operator
- Find $[T]$ with respect to the standard basis $\{1, x, x^2\}$ and show that $\ker(T) = \{0\}$
- Use the above result to find the equation of the parabola passing through the points, $(-3, 6)$, $(2, -3)$, $(5, 4)$

42) Let A be any matrix and prove that

- $\text{null}(A^T A) = \text{null}(A)$

b. $\text{null}(AA^T) = \text{null}(A^T)$

Hint: Use the orthogonality of the fundamental subspaces. To prove set equality show that they are both subsets of each other: $S \subseteq T$ iff $x \in S \Rightarrow x \in T$

43) Let P be a (not necessarily orthogonal) projection.

- a. Prove that for any $\mathbf{y} \in \text{range}(P)$, $P\mathbf{y} = \mathbf{y}$.
- b. Prove that $\text{range}(P) = \text{null}(I - P)$.
- c. Prove that if P is full rank, then it must be the identity.

44) Let V be a finite dimensional vector space and let P be an orthogonal projection onto the subspace W . Show that $\|P\mathbf{x}\| \leq \|\mathbf{x}\|$ for all $\mathbf{x} \in V$. (Such a map is called a contraction)

45) Prove the converse, that is if P is a projection onto W that is also a contraction, then P is an orthogonal projection onto W .